Project Title:

Building a Kali Linux Digital Forensics Lab for Evidence Collection and Analysis

1. Introduction

In this project, I will be setting up a digital forensics environment using Kali Linux to gain hands-on experience with real forensic tools and techniques. The goal is to understand how digital evidence is collected, analyzed, and managed in a secure and professional way. This project will also help me get familiar with tools that are commonly used in cybersecurity and forensic investigations.

The overall scope includes installing Kali Linux in a virtual machine, identifying important forensic tools, and explaining how those tools can be used to acquire and manage evidence while following proper forensic procedures.

2. Kali Linux Installation Plan

2.1 Installation Method

For this project, I will be using a virtual machine (VirtualBox) to run Kali Linux on my personal laptop. I already have experience working with virtual machines, so this setup will allow me to easily manage and test different tools without affecting my main system.

2.2 Justification

I chose a virtual machine instead of a cloud option like Azure because it gives me more control over the environment and makes it easier to practice locally. It also allows me to take snapshots, which is useful if something breaks or if I want to go back to a clean state. This setup fits my current workflow and makes it easier to focus on learning the tools.

2.3 Pre-installation Steps

Before installing Kali Linux, I will complete the following steps:

- Make sure my laptop meets the system requirements (RAM, storage, CPU)
- Download and install Oracle VirtualBox
- Download the official Kali Linux ISO file
- Create a new virtual machine and allocate resources (RAM, CPU, storage)
- Attach the Kali Linux ISO file to the VM

2.4 Post-installation Configuration

After installing Kali Linux, I will:

- Update the system using apt update and apt upgrade
- Set up user accounts and basic system settings

- Check that networking is working properly
- Install or verify forensic tools inside Kali
- Take a snapshot of the clean system so I can always return to it if needed

3. Forensic Tools Identification

3.1 Disk Imaging Tools

- **FTK Imager**

  This tool is used to create exact copies (images) of storage devices. It helps preserve the original evidence so it can be analyzed without being changed.

- **dc3dd**

  A command-line tool used for disk imaging that also allows hashing and logging. This helps verify that the data has not been altered during the imaging process.

- **Autopsy**

  A forensic analysis tool that helps examine disk images, recover deleted files, and analyze system activity. It provides a user-friendly interface for investigations.

3.2 Live Acquisition Tools

- **dd**

  A basic command-line tool used to copy data from one location to another, often used in forensic imaging.

- **LiME (Linux Memory Extractor)**

  Used to capture memory from a running system. This is important because memory contains active processes and temporary data that can be useful in an investigation.

3.3 Memory Analysis Tools

- **Volatility**

  One of the most important tools for analyzing memory dumps. It can show running processes, network connections, and signs of malware.

- **Rekall**

  Another memory analysis tool that is fast and useful for incident response and forensic investigations.

3.4 Network Analysis Tools

- **Wireshark**

  A tool used to capture and analyze network traffic in real time. It helps identify suspicious activity or data transfers.

- **tcpdump**

  A command-line tool used for capturing network packets. It’s useful for quick and lightweight network analysis.

3.5 Malware Analysis Tools

- **Ghidra**

  A reverse engineering tool used to analyze malware and understand how it works.

- **Radare2**

  A powerful framework used for analyzing binaries and performing reverse engineering.

3.6 Steganography Tools

- **Steghide**

  Used to detect and extract hidden data inside files like images or audio.

- **zsteg**

  A tool used to find hidden data in image files, especially PNG and BMP formats.

3.7 Rationale

I selected these tools because they are widely used in the cybersecurity and forensics field and are available in Kali Linux. Tools like Autopsy and Volatility are essential for deep analysis, while Wireshark and tcpdump help with network investigations. Together, these tools cover different areas of digital forensics and allow me to get a full understanding of how investigations are performed.

4. Evidence Acquisition and Management Strategy

4.1 High-Level Overview

The process of acquiring digital evidence will involve creating forensic images of storage devices and capturing memory when needed. These images will then be analyzed using the tools listed above to identify important data, suspicious activity, or possible threats.

4.2 Best Practices

During this process, I will follow important forensic best practices, including:

- Maintaining a clear chain of custody
- Avoiding changes to original evidence
- Using hashing (MD5/SHA256) to verify data integrity
- Documenting every step taken during the investigation
- Keeping evidence stored securely

5. Conclusion

This proposal outlines my plan to set up a Kali Linux-based forensic lab and use real tools to understand how digital investigations work. By following a structured installation process, selecting the right tools, and applying proper forensic methods, I am prepared to move forward with the project and begin working in a hands-on environment that reflects real-world forensic practices.